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第1章 序論

1.1 Introduction

In recent years, Human-Robot Collaboration (HRC) has attracted significant attention as an important research domain in which humans and robots cooperate to accomplish tasks in diverse working environments. The aim is not only for robots to interpret human intentions more accurately but also to enable smooth and intuitive communication. When working in shared spaces, the synergy between humans and robots can be expected to improve both efficiency and safety.

One of the major challenges in HRC lies in the discrepancy between human intention and robotic perception. Such mismatches often cause reductions in task efficiency and increase the difficulty of operation. In particular, operational mismatches stemming from the differences in embodiment between humans and robots represent a fundamental bottleneck in collaborative work. To address this issue, approaches that leverage Mixed Reality (MR) to intuitively map human operations onto robots have been actively studied. MR enables users to recognize robotic motion as an extension of their own body, thereby fostering more natural and seamless collaboration [?].

Frameworks for intuitive operation support in HRC have also been reported not only with MR but also through Augmented Reality (AR) and Virtual Reality (VR). For example, VR-based methods have been proposed for imitation learning and motion data collection, allowing robots to efficiently acquire manipulation skills [?]. Meanwhile, AR-based approaches have sought to present robot states and task procedures to the user, thereby enhancing task comprehension [?]. Although these works have shown the effectiveness of bridging recognition gaps between humans and robots, limitations remain in terms of immersion and interaction with the real-world environment. In this context, MR is particularly promising because it allows for the seamless integration of real-world environments with virtual information, thus enabling more intuitive and adaptive HRC.

A representative prior study is the work by Esaki et al., entitled *Immersive Robot Teleoperation Based on User Gestures in Mixed Reality Space (IRT-MRO)* [?]. IRT-MRO demonstrated the effectiveness of HRC using MR by enabling humans to intuitively manipulate virtual objects in MR space, with the robot recognizing and executing corresponding positional and postural changes. However, several challenges remain unresolved, including the integration of environmental perception, the assurance of stable robot posture, and the autonomy of base relocation strategies.

To overcome these limitations, this study proposes a new framework: **Human–Robot Interaction with Correction and Suggestion (HRI-CAS)**.

The proposed system integrates the following three modules:

1. **Target Bottle Inference (TBI)**: infers the intended object of manipulation based on user features.
2. **Base Relocation based on Inverse Reachability Map (BRI)**: determines base relocation strategies using IRM.
3. **Human-Intent Adaptive Grasp Manipulation (HIGM)**: compensates for differences between human intentions and robotic embodiment during grasping tasks.

By combining these components, the proposed framework enables stable collaborative task execution that does not rely on user expertise, aiming for bidirectional and interactive HRC through MR space.

Contributions

The main contributions of this study can be summarized as follows:

- **Proposal of the Target Bottle Inference (TBI) module** By integrating user operation features such as hand proximity, contact, gaze, and arm motion, and weighting them according to reachability, the TBI module infers the intended object of manipulation. This approach suppresses unnecessary information presentation while supporting intuitive target selection.
- **Realization of the Base Relocation strategy based on the Inverse Reachability Map (BRI)** By utilizing the Inverse Reachability Map, the system computes feasible base positions for grasping and visualizes candidate paths in MR space. Users can edit these paths interactively, thereby achieving a balance between autonomy and interactivity in determining whether base relocation is necessary.
- **Implementation of the Human-Intent Adaptive Grasp Manipulation (HIGM) framework** By combining pregrasp generation, nullspace optimization, and a finite state machine (FSM), the system continuously harmonizes user operations with robot manipulability. This enables seamless collaborative manipulation based on mutual correction and suggestion.

Through these contributions, the proposed system allows robots to respect user intentions while providing appropriate correction and suggestions, ensuring stable task execution regardless of user expertise.

Furthermore, the proposed HRI-CAS framework demonstrates applicability to real-world tasks where humans and robots closely collaborate, such as in manufacturing sites, remote maintenance, and medical support. This work is thus expected to contribute to the further development of HRC research.

1.2 Related Work

Robots are expected to support human life and activities in a wide range of fields such as medicine, manufacturing, space exploration, and agriculture. However, many challenges remain for humans and robots to work collaboratively in the same space. One of these is the lack of mutual understanding of intentions and actions. Humans often find it difficult to accurately predict the capabilities and behaviors of robots, while robots cannot fully interpret humans' high-level goals. As a result, "gaps of execution" and "gaps of evaluation" arise, hindering smooth collaboration. Conventional interfaces have mainly relied on joysticks or keyboards, but visualization on 2D displays provides insufficient depth information, limiting the operator's spatial awareness and sense of immersion.

To address these challenges, research on Human-Robot Collaboration (HRC) using Mixed Reality (MR) has made significant progress in recent years. By overlaying information about robot states and environmental conditions in MR space, users can perform more intuitive operations, and improvements in efficiency for teleoperation and collaborative tasks can be expected [?, ?]. In addition, many studies have explored applications of AR and VR. In VR, methods have been proposed to support skill acquisition by robots through imitation learning and motion data collection [?], while in AR, attempts have been made to assist workers' understanding by presenting task procedures and robot states [?]. Nevertheless, these approaches still face limitations in interaction with the real environment and in providing a sense of immersion. In this context, MR, which can integrate the real world with virtual information, is receiving particular attention as a promising foundation technology for HRC.

Previous MR-based HRC research can largely be classified into two directions.

First, research focusing on **visualization of robot states**. This direction projects sensor data and robot internal states into MR space, thereby supporting users' situational awareness and decision-making [?]. While this approach contributes to enhancing environmental understanding and reducing cognitive load, direct intervention in robot motion planning and control remains limited.

Second, research focusing on **intuitive mapping of user operations**. By transforming hand tracking and gesture input into robot motions, these approaches enhance operability and immersion.

In this category, the work of Esaki et al. is representative. They proposed the *Immersive Robot Teleoperation Based on User Gestures in Mixed Reality Space (IRT-MRO)* [?]. In IRT-MRO, objects are placed in MR space, and users directly manipulate their positions and orientations. The manipulated results are transferred to the robot side, enabling the robot to execute the same operations in the physical world. This demonstrated the effectiveness of MR in improving the intuitiveness of robot teleoperation.

However, IRT-MRO also leaves several issues unresolved. First, because the system simply transfers the object manipulation performed by the user in MR space, it does not take into account whether the robot can actually reach the target object or grasp it stably.

Second, the posture of the robot arm is not always guaranteed to remain safe, and problems such as singularity or joint limit violations can occur.

Third, when an object is located outside the reachable area of the robot, the system does not provide an appropriate relocation strategy for the robot base, leaving the operation incomplete.

In summary, previous studies have clarified the potential of MR as an intuitive interface for HRC. Nevertheless, they still face challenges regarding (1) robust target inference, (2) ensuring robot posture and grasp stability, and (3) achieving autonomous yet interactive base relocation strategies. These issues become bottlenecks in realizing seamless human-robot cooperation in dynamic and complex environments.

To address these challenges, this paper proposes a new framework called **Human–Robot Interaction with Correction and Suggestion (HRI-CAS)**. Unlike previous MR-based systems, HRI-CAS integrates:

1. **Target Bottle Inference (TBI)**: a module that infers the user’s intended target based on multimodal features such as gaze, head direction, and hand motion, with adaptive information-level control.
2. **Base Relocation based on Inverse Reachability Map (BRI)**: a module that determines optimal robot base positions when the target is unreachable, by leveraging IRM-based optimization and MR-based user corrections.
3. **Human-Intent Adaptive Grasp Manipulation (HIGM)**: a module that integrates Pregrasp design, nullspace optimization, and finite state machines to adaptively align user intention with the robot’s kinematic constraints.

Through these three modules, HRI-CAS enhances the robustness of target selection, guarantees stable robot motion, and provides a cooperative relocation mechanism, thereby advancing MR-based HRC beyond the limitations of IRT-MRO.

1.3 Human–Robot Interaction with Correction and Suggestion (HRI-CAS)

Human–Robot Interaction with Correction and Suggestion (HRI-CAS) addresses several limitations observed in conventional MR-based HRC systems, including insufficient utilization of environmental information, lack of stable posture guarantees, and inadequate autonomy in base relocation strategies.

This system is built upon the foundation of *Immersive Robot Teleoperation Based on User Gestures in Mixed Reality Space (IRT-MRO)* and improves upon the issues that remained in IRT-MRO. Whereas IRT-MRO employed a Unity-based holographic application on the HoloLens 2, the proposed system adopts a Unity-based passthrough application using the Meta Quest. Although both devices are capable of projecting real-world objects into MR space, they differ significantly in their display methods.

The HoloLens 2 adopts an optical see-through display, allowing the user to perceive both the real environment and virtual objects through a transparent screen. However, its limited field of view and restricted brightness have been widely noted. In contrast, as illustrated in Figure1, the Meta Quest employs a passthrough display system, which projects images captured by external cameras onto the display, enabling a wide field of view and high-resolution rendering. This provides the user with a more natural and immersive MR experience, enhancing both environmental understanding and intuitive operation.

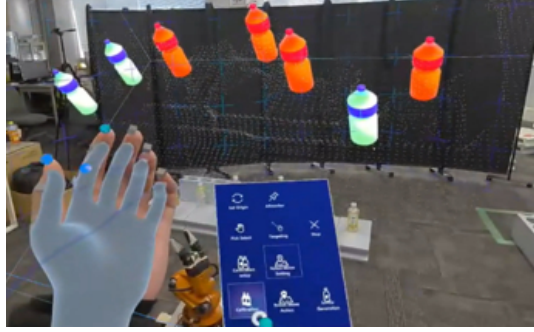


Fig.1.1: Figure1

1.3.1 Implementation Platform and System

In this study, the experimental platform adopted was the **KUKA YouBot**. The robot is a mobile manipulator equipped with an omnidirectional base, a 5-DOF manipulator, and a two-finger gripper. Within the proposed system, changes in the position and orientation of target objects detected in MR space, as well as user manipulations, are reflected directly in the robot's motion. This ensures smooth and seamless interaction between the human and the robot. Furthermore, the robot's behavior is immediately fed back into MR space, providing the operator with precise visual information in real time.

The graphical architecture of the system is illustrated in Figure 2 and is composed of the following three modules:

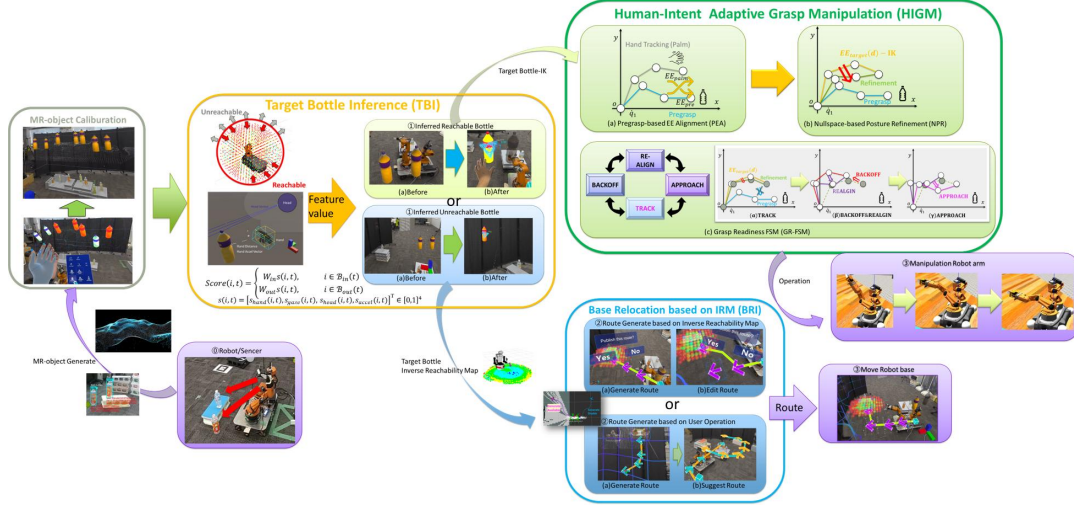


Fig.1.2: Figure2

Target Bottle Inference (TBI) module: This module tracks the user's gaze, head orientation, and hand motion within MR space, extracting geometric relationships with candidate bottles as features. These multimodal features are then processed through an integrated scoring framework to infer the bottle that the user intends to manipulate. By jointly evaluating diverse behavioral cues, the TBI module enables highly accurate and real-time target selection.

Base Relocation based on Inverse Reachability Map (BRI) module: When the target bottle lies outside the robot's reachable workspace, this module becomes active. First, an Inverse Reachability Map (IRM) is generated based on the Reachability Map (RM), and candidate base positions that make the target reachable are extracted. Next, an optimization problem is solved considering both the IRM score and the movement cost, yielding the optimal base configuration. Additionally, candidate paths are visualized within MR space, allowing users to refine or adjust the proposed routes interactively.

Human-Intent Adaptive Grasp Manipulation (HIGM) module: This module aims to reconcile the differences between the user's intended manipulation and the robot's physical and kinematic constraints. Specifically, it integrates reference postures obtained from hand-tracking information with robot-preferred pregrasp postures, and applies nullspace optimization to adjust joint configurations. As a result, the user can achieve stable grasping with natural operational feedback, without needing to be aware of the robot's inherent kinematic constraints.

Through the integration of these modules, the proposed system achieves adaptive and intuitive motion planning that was not sufficiently addressed in IRT-MRO. The robot is

now able to select actions according to the user's intended target, while providing adaptive grasp correction without requiring the user to account for robot-specific constraints. This significantly reduces dependency on the user's operational expertise and ensures smooth, stable operation even for novice users.

1.3.2 MR System Implementation

The MR system is implemented on **Unity** and **MRTK3 (Mixed Reality Toolkit)**, in compliance with the OpenXR standard. The user can manipulate MR objects directly through hand gestures, offering an intuitive and natural input interface. Moreover, by presenting the robot's behavioral intentions through MR objects, the system improves the operator's understanding of the robot's decision-making process, thereby enhancing human–robot collaboration.

1.4 Module of Human–Robot Interaction with Correction and Suggestion (HRI-CAS)

1.4.1 Target Bottle Inference (TBI) Module

Definition of Features: The TBI module aims to estimate the user's intended target object based on bodily movements. Figure 3 illustrates the distribution of operational features in the information space.

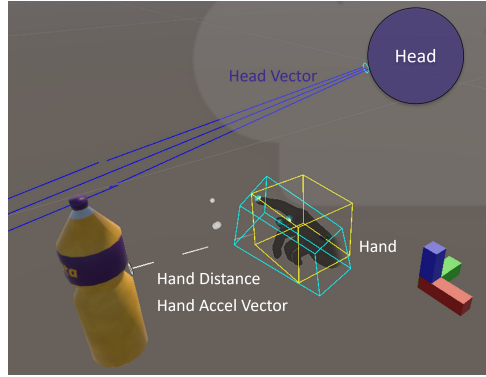


Fig.1.3: Figure3

Let the position of a target object o_i be denoted as $\mathbf{p}_i(t)$, and the positions of the user's eyes, head, and hand be $\mathbf{p}_{eye}(t)$, $\mathbf{p}_{head}(t)$, and $\mathbf{p}_{hand}(t)$, respectively. For each object, the normalized direction vectors from these body parts are defined as:

$$\mathbf{d}_i^{eye}(t) = \frac{\mathbf{p}_i(t) - \mathbf{p}_{eye}(t)}{\|\mathbf{p}_i(t) - \mathbf{p}_{eye}(t)\|}, \quad \mathbf{d}_i^{head}(t) = \frac{\mathbf{p}_i(t) - \mathbf{p}_{head}(t)}{\|\mathbf{p}_i(t) - \mathbf{p}_{head}(t)\|}, \quad \mathbf{d}_i^{hand}(t) = \frac{\mathbf{p}_i(t) - \mathbf{p}_{hand}(t)}{\|\mathbf{p}_i(t) - \mathbf{p}_{hand}(t)\|}. \quad (1.1)$$

These vectors geometrically define the directions in which the user is focusing attention or exerting effort. To incorporate dynamics of motion, normalized gaze, head orientation, and hand acceleration vectors are introduced as $\hat{\mathbf{v}}_{gaze}(t)$, $\hat{\mathbf{v}}_{head}(t)$, and $\hat{\mathbf{a}}_{hand}(t)$.

Cosine Similarity Scores: To evaluate whether the user's body movements align with the direction of candidate objects, cosine similarity between these features and the direction vectors is computed:

$$s_{\text{gaze}}(i, t) = \max(0, \hat{\mathbf{v}}_{\text{gaze}}(t) \cdot \mathbf{d}_i^{\text{eye}}(t)), \quad (1.2)$$

$$s_{\text{head}}(i, t) = \max(0, \hat{\mathbf{v}}_{\text{head}}(t) \cdot \mathbf{d}_i^{\text{head}}(t)), \quad (1.3)$$

$$s_{\text{accel}}(i, t) = \max(0, \hat{\mathbf{a}}_{\text{hand}}(t) \cdot \mathbf{d}_i^{\text{hand}}(t)). \quad (1.4)$$

Here, s_{gaze} represents the degree to which the gaze is directed toward each object, s_{head} represents the degree to which the head orientation captures each object, and s_{accel} represents the degree to which the hand acceleration is aligned with the object's direction.

In addition, physical graspability is considered by introducing: (i) a proximity score s_{prox} based on the Euclidean distance between the hand and the object, and (ii) a binary contact variable $c_i(t)$:

$$s_{\text{prox}}(i, t) = \text{clip}\left(1 - \frac{\|\mathbf{p}_{\text{hand}}(t) - \mathbf{p}_i(t)\|}{R_{\text{max}}}, 0, 1\right), \quad c_i(t) \in \{0, 1\}. \quad (1.5)$$

Thus, the module integrates both body movement and physical graspability.

Score Integration: The feature vector is defined as

$$\mathbf{s}(i, t) = [s_{\text{prox}}, c_i, s_{\text{gaze}}, s_{\text{head}}, s_{\text{accel}}]^\top, \quad (1.6)$$

and the integrated score is calculated with different weightings depending on whether the target belongs to the reachable set $\mathcal{B}_{\text{in}}(t)$ or the unreachable set $\mathcal{B}_{\text{out}}(t)$:

$$S(i, t) = \begin{cases} \mathbf{w}_{\text{in}}^\top \mathbf{s}(i, t), & i \in \mathcal{B}_{\text{in}}(t), \\ \mathbf{w}_{\text{out}}^\top \mathbf{s}(i, t), & i \in \mathcal{B}_{\text{out}}(t). \end{cases} \quad (1.7)$$

Here, $\mathbf{w}_{\text{in}}$ and $\mathbf{w}_{\text{out}}$ are predefined weight vectors. For reachable objects, emphasis is placed on contact and proximity, ensuring robustness against noisy gaze fluctuations. For unreachable objects, emphasis is placed on gaze and hand motion, capturing the user's distal intention. This design enables flexible and error-tolerant inference depending on the situation.

Temporal Smoothing: To suppress rapid fluctuations caused by momentary body movements, an exponential moving average is applied:

$$\bar{S}(i, t) = \alpha S(i, t) + (1 - \alpha) \bar{S}(i, t - 1), \quad \alpha \in (0, 1]. \quad (1.8)$$

Here, $\bar{S}(i, t)$ is the smoothed integrated score, and α is the smoothing coefficient. This prevents transient misdetections and allows stable inference of user intention.

Final Decision Rule: The final target object is selected as the one maximizing the smoothed score over the entire candidate set $\mathcal{B}$:

$$i^*(t) = \arg \max_{i \in \mathcal{B}} \bar{S}(i, t). \quad (1.9)$$

However, the selected object must satisfy threshold conditions:

$$\bar{S}(i^*(t), t) \geq \begin{cases} T_{\text{in}}, & i^*(t) \in \mathcal{B}_{\text{in}}(t), \\ T_{\text{out}}, & i^*(t) \in \mathcal{B}_{\text{out}}(t). \end{cases} \quad (1.10)$$

Here, T_{in} and T_{out} are thresholds for the reachable and unreachable sets, respectively. This two-step decision ensures that only targets with sufficiently strong intentional signals are confirmed.

Information Presentation Adjustment: When presenting inference results to the user, the TBI module adjusts the amount of information to prevent cognitive overload while supporting intuitive understanding.

First, as shown in Figure 4, each candidate bottle is classified by the *Reachability Map* (RM) as “graspable” or “non-graspable.” Graspable objects are displayed with a base color, while non-graspable ones are rendered semi-transparent with reduced saturation, ensuring they do not unnecessarily attract the user’s attention.



Fig.1.4: Figure4

Furthermore, as illustrated in Figure 5, once the target is inferred, all non-selected objects are displayed in semi-transparent form. At this stage, different highlighting schemes are used for graspable and non-graspable objects: graspable targets are emphasized with a color indicating manipulability, while non-graspable targets are highlighted with a different tone to indicate the need for base relocation.

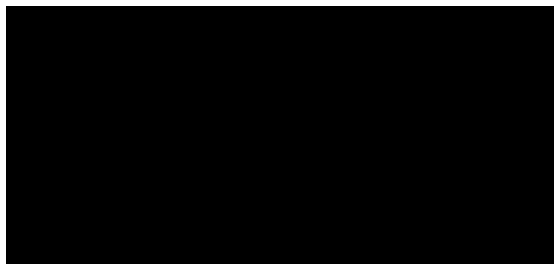


Fig.1.5: Figure4

Through this two-step information adjustment, users can clearly distinguish between “objects that can currently be grasped” and “objects that require relocation.” By emphasizing the final inferred target, the system reduces selection errors and alleviates cognitive load, supporting intuitive decision-making.

Summary of TBI In summary, the TBI module geometrically defines multiple bodily features such as gaze, head orientation, and hand motion, and integrates them through cosine similarity, proximity, and contact indicators to construct a robust scoring framework. Temporal smoothing suppresses transient misdetections, ensuring stable real-time inference.

Furthermore, the inference results are visually presented in MR space:

1. Objects are classified by the Reachability Map (RM) as graspable or non-graspable, displayed either in base color or semi-transparent with reduced saturation.
2. The inferred target bottle is emphasized, with different highlight colors used for graspable and non-graspable objects, providing explicit feedback to the user.

By jointly addressing “target inference” and “information presentation,” the TBI module reduces cognitive load, prevents misselection, and establishes accurate recognition of human intention as the foundation of the HRI-CAS framework.

1.4.2 Base Relocation based on Inverse Reachability Map (BRI)

When the target lies outside the robot’s reachable workspace, the robot base itself must be relocated. Naively exploring base candidates leads to prohibitive computational cost and makes it difficult to obtain configurations that respect human–robot collaboration. Therefore, as illustrated in Fig. 6, we introduce two concepts widely used in manipulator research—the *Reachability Map (RM)* and the *Inverse Reachability Map (IRM)*—to systematically generate and evaluate candidate base poses [?].

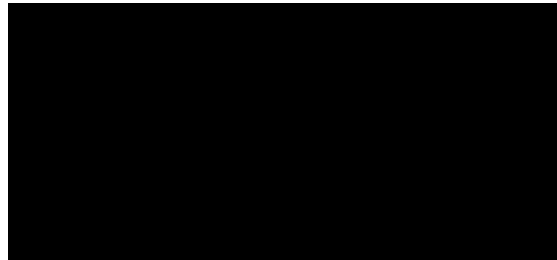


Fig.1.6: Figure6

Basics of reachability and inverse reachability: Given a robot base pose $\mathbf{b}$, the workspace that the end-effector (EE) can reach from $\mathbf{b}$ is defined as

$$RM(\mathbf{b}) = \{\mathbf{p} \in \mathbb{R}^3 \mid \exists \theta \text{ s.t. } FK(\theta, \mathbf{b}) = \mathbf{p}\}, \quad (1.11)$$

where $FK(\theta, \mathbf{b})$ denotes forward kinematics. In other words, if there exists a joint configuration θ , the point $\mathbf{p}$ is reachable from the fixed base $\mathbf{b}$. Thus, the RM specifies “how far the arm can reach from a fixed base pose.”

Conversely, for a given task point $\mathbf{p}_t$, the set of base poses that makes $\mathbf{p}_t$ reachable is

$$\mathcal{B}(\mathbf{p}_t) = \{\mathbf{b} \in \mathbb{R}^3 \mid \mathbf{p}_t \in RM(\mathbf{b})\}, \quad (1.12)$$

which we call the *Inverse Reachability Map (IRM)*. IRM is the inverse mapping that provides “the set of base poses from which the task point is reachable,” and is essential when base relocation is considered [?].

Beyond a binary reachable/unreachable decision, RM can evaluate stability/redundancy by counting the number of inverse kinematic (IK) solutions available at a point. In this work, the RM score for position $\mathbf{p}$ is defined by

$$S_{RM}(\mathbf{p}) = N_{IK}(\mathbf{p}). \quad (1.13)$$

A larger number of solutions implies greater redundancy and flexibility in motion, which typically increases grasp success rates.

Since IRM is the inverse of RM, each candidate base pose $\mathbf{b} \in \mathcal{B}(\mathbf{p}_t)$ for target $\mathbf{p}_t$ inherits the corresponding RM score:

$$S_{IRM}(\mathbf{b}; \mathbf{p}_t) = N_{IK}(\mathbf{p}_t \mid \mathbf{b}). \quad (1.14)$$

This allows us to assess candidate base configurations not only by “reachability,” but also by “how redundantly and stably” the target can be reached.

Optimization problem: We now select the optimal base pose from the candidate set $\mathcal{B}(\mathbf{p}_t)$. As shown conceptually in Fig. 7, we formulate two cases depending on the availability of environment/navigation information.

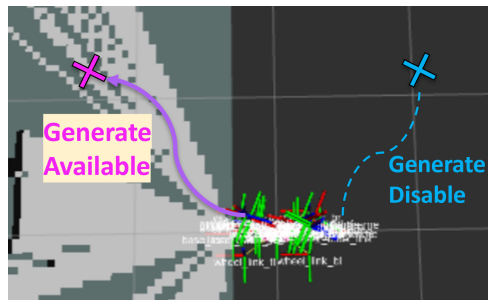


Fig.1.7: Figure7

Case: Known path If a navigable path to the goal exists, we select the base by jointly considering the IRM score $S_{IRM}(\mathbf{b}; \mathbf{p}_t)$ and the movement cost $C(\mathbf{b})$:

$$\mathbf{b}^* = \arg \max_{\mathbf{b} \in \mathcal{B}(\mathbf{p}_t)} [S_{IRM}(\mathbf{b}; \mathbf{p}_t) - \lambda C(\mathbf{b})]. \quad (1.15)$$

Here, $C(\mathbf{b})$ denotes the cost to move from the current base $\mathbf{b}_0$ to $\mathbf{b}$ (e.g., Euclidean distance or shortest path length on an occupancy grid). The parameter λ balances “reach stability” versus “movement efficiency.” The resulting candidate base poses and waypoint sequences are visualized in MR space; the user can then modify the path by directly manipulating the visualized waypoints. This mechanism realizes the coexistence of autonomous planning and user interaction, as depicted in Fig. 8.

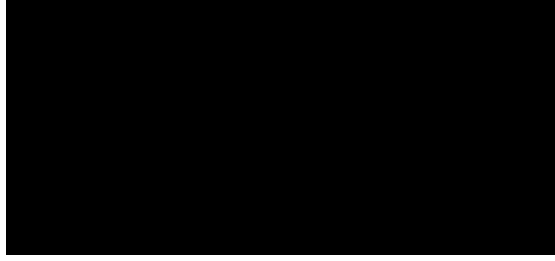


Fig.1.8: Figure8

Case: Unknown path When environment information is incomplete and path generation fails, we adopt the IRT-MRO [?] scheme: the user directly lays out a base route in MR space. On top of this, we introduce a local autonomous refinement using a cost map. As shown in Fig. 9, each waypoint is adjusted to a safe and feasible pose that accounts for obstacles and environmental changes. The refined route is immediately reflected in MR space so that the user can visually grasp the autonomous corrections made by the robot.

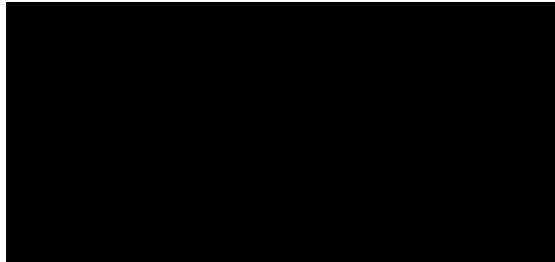


Fig.1.9: Figure4

Summary of BRI In summary, our base relocation strategy adopts a two-track structure: (i) when a path is known, we perform autonomous optimization based on the IRM score and movement cost; (ii) when a path is unknown, we combine user-specified routes with cost-map-based refinement. This design satisfies both robot autonomy and user-driven interaction, achieving flexible and robust base relocation.

1.4.3 Human-Intent Adaptive Grasp Manipulation (HIGM)

Even when the base position is appropriate, discrepancies between human intent and the robot’s kinematic constraints can cause the end-effector (EE) to approach the target in unstable poses. This can lead to grasp failure or excessive joint load. To address this issue, we propose an *adaptive grasp framework* that integrates Pregrasp design, null-space optimization, and a finite state machine (FSM).

Pregrasp and EE interpolation: Human operation alone may command poses that are difficult for the robot to execute. For example, if the user attempts to pinch the top of a bottle, directly mapping the hand-back pose to the EE could result in an unstable grasp configuration.

To address this, as shown in Fig. 10, we generate a robot-friendly reference pose (Pregrasp pose $\mathbf{x}_{pre}$) based on the target position $\mathbf{p}_t$. The actual EE target is defined by interpolating between the user’s palm pose EE_{palm} and the Pregrasp pose EE_{pre} , depending on the distance d to the target:

$$EE_{target}(d) = (1 - \beta(d)) EE_{palm} + \beta(d) EE_{pre}, \quad (1.16)$$

where $\beta(d) \in [0, 1]$ is a distance-dependent interpolation coefficient.

With this scheme, user intent (EE_{palm}) dominates when the target is far away, and as the robot approaches, the weight gradually shifts toward the stable Pregrasp pose (EE_{pre}). In other words, the user’s motion is “magnetically guided” toward the object—similar to aim assist in video games—preserving intuitive control while ensuring stable and safe grasping for the robot.

Furthermore, Pregrasp interpolation provides the global reference, while null-space optimization ensures local stability and safety. Thus, interpolation defines “which pose should be aimed for,” and null-space optimization ensures “how to safely approach that pose.” Combining them achieves grasping that respects user intent while adapting to the robot’s kinematic limits.

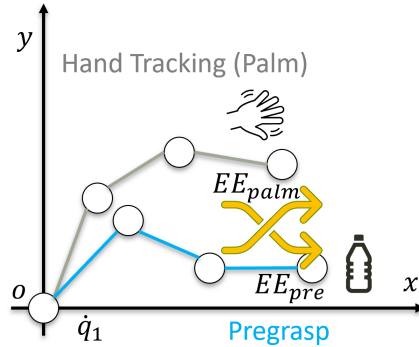


Fig.1.10: Figure10

Null-space optimization: For redundant manipulators, multiple joint configurations can realize the same EE pose. The set of joint motions that do not affect the EE is called the *null space* [?].

Formally, the task-space velocity $\dot{x}$ and joint velocity $\dot{\theta}$ satisfy

$$\dot{x} = J\dot{\theta}. \quad (1.17)$$

A velocity $\dot{\theta}_{null}$ belongs to the null space if

$$J\dot{\theta}_{null} = 0. \quad (1.18)$$

Such motions leave the EE unaffected while adjusting only the joint posture.

We exploit this property to simultaneously achieve the primary task and optimize joint posture. The joint velocity is given by

$$\dot{\theta} = J^+ \dot{x}_d + P \nabla_{\theta} H(\theta), \quad (1.19)$$

where the damped least-squares pseudoinverse is

$$J^+ = J^{\top} (J J^{\top} + \lambda I)^{-1}, \quad (1.20)$$

and $P = I - J^+ J$ is the null-space projector. The first term guarantees EE tracking, faithfully reproducing the user's hand motion. The second term optimizes posture in the null space using the merit function $H(\theta)$ (Fig. 10):

$$H(\theta) = w_{mani} \log \det(J J^{\top}) + w_{limit} \sum_i \log m_i(\theta) - \gamma \|\theta - \theta_{pre}\|^2. \quad (1.21)$$

The roles of each term are:

- $w_{mani} \log \det(J J^{\top})$: maximizes Yoshikawa's manipulability, avoiding singularities and preventing the arm from losing freedom. This maintains motion redundancy, providing the user with smooth and continuous operation.
- $w_{limit} \sum_i \log m_i(\theta)$: enlarges the margin $m_i(\theta)$ to joint limits, preventing joints from approaching boundaries. This ensures mechanical safety and reduces the risk of sudden motion stops, improving user comfort.
- $-\gamma \|\theta - \theta_{pre}\|^2$: maintains consistency with the Pregrasp pose, preventing optimization from drifting too far from user intent. This balances intuitive operation and kinematic stability.

Thus, null-space optimization ensures (1) stability through singularity avoidance, (2) safety through joint-limit avoidance, and (3) intent preservation through Pregrasp consistency. It functions not only as numerical stabilization, but as a foundation linking “robotic physical constraints” with “the user's continuous intuitive experience.”

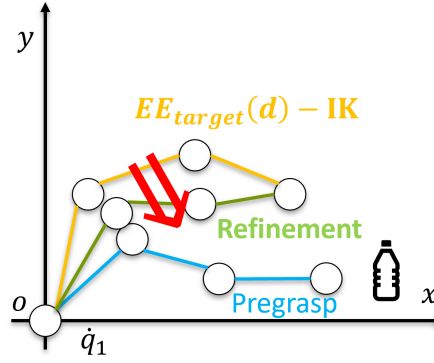


Fig.1.11: Figure10

Finite State Machine (FSM): To manage grasp execution safely and reliably, we introduce a four-state FSM [?]. The FSM ensures both user-intent reflection and robot safety, preventing failures such as entering the target in unstable postures.

The roles and transitions of the states are:

1. **TRACK:** The robot follows the target using Palm/Pregrasp interpolation. Mathematically, the EE target pose $EE_{target}(d)$ is generated and joint angles are updated via DLS-IK. User experience: “the robot hand smoothly follows my hand.”
2. **BACKOFF:** If manipulability $M(\theta) < \tau_{mani}$ or margin $m_i(\theta) < \tau_{limit}$, the robot retreats to avoid instability. This prevents “robot jamming” and provides the user with a sense of safety.
3. **REALIGN:** After BACKOFF, null-space optimization realigns the joints until $M(\theta) \geq \tau_{mani}$ and $m_i(\theta) \geq \tau_{limit}$. User experience: “the robot autonomously reconfigures itself,” revealing its adaptiveness.
4. **APPROACH:** Once stability is secured and the target distance $d < \tau_{dist}$, the EE approaches to grasp. This corresponds to “the robot decisively grabs,” improving grasp success.

The FSM links mathematical criteria (manipulability, joint margin, distance thresholds) with user experience (tracking feel, safety, adaptiveness, visible grasp action). In particular, the BACKOFF $\rightarrow$ REALIGN transition guarantees that even if the user makes errors, the robot autonomously corrects them, realizing “collaborative safety” in HRC.

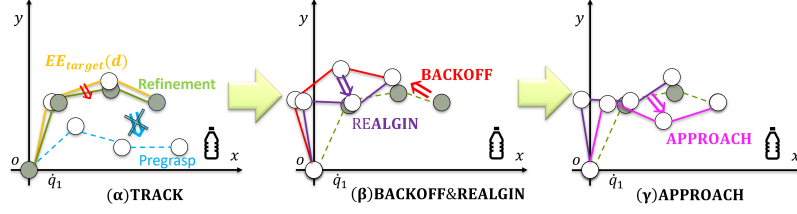


Fig.1.12: Figure10

1.4.4 Significance of Module Integration

The proposed HRI-CAS consists of three modules: Target Bottle Inference (TBI), Base Relocation based on IRM (BRI), and Human-Intent Adaptive Grasp Manipulation (HIGM). Each module functions independently, but more importantly, they complement each other to overcome the limitations faced by conventional MR-based HRC systems.

First, TBI accurately estimates the user's intent, clarifying "which object should be manipulated." This object information is then passed to BRI, which determines whether base relocation is necessary depending on the object's reachability. Subsequently, HIGM leverages the outputs of TBI and BRI to compensate for discrepancies between human intent and robot kinematic constraints, ensuring stable grasping.

Through the integration of TBI, BRI, and HIGM, the system establishes a coherent workflow: (1) recognizing user intent with high precision, (2) ensuring reachability through appropriate base relocation, and (3) guaranteeing grasp stability. As a result, the robot operates intuitively and adaptively without over-relying on the user's physicality, providing a stable interaction experience for both novice and expert users.

Thus, the significance of this research lies not merely in individual technical improvements, but in the establishment of an integrated framework for HRC in MR environments, encompassing the flow of "intent recognition $\rightarrow$ reachability assurance $\rightarrow$ stable grasp."

参考文献

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